

Minutes of Meeting: DASS Client Meet 4th March

March 7, 2026

Meeting Details

- **Date:** March 4th, 2026
- **Time:** 4:00 PM
- **Platform:** Online Meeting(Gmeet)

Attendees

- **Client Team:** Manish Ranjan, Madhu Nyalam
- **Student Team:** Venya, Amish, Keshav, Advait, Vikesh

Meeting Agenda

Discussion on context optimization, transition to Retrieval-Augmented Generation (RAG) for large medical records, and model performance evaluation.

1 Context & Token Optimization

- **Growing Context:** The team discussed the challenge of expanding context as chat history grows, especially when integrating several years of health data.
- **Data Type Focus:** While there was a brief discussion about multimodal LLMs for processing pictures, the client decided to focus strictly on text for the current phase since all their medical records are currently being digitized into text.
- **Optimization Goal:** The next priority is to optimize how the model handles growing context across a chat session to maintain performance and cost-efficiency.

2 Technical Architecture: Re-evaluating RAG

- **Shift in Strategy:** Although RAG was previously deemed as extra by the client, the student team proposed reconsidering it due to the scale of Electronic Medical Records (EMR).
- **Scalability:** A single JSON record was noted to be around 65,000 characters (approx. 10k–14k tokens). Sending the entire record with every query would exhaust token limits and increase costs.
- **Proposed RAG Mechanism:**

- Convert EMR and user queries into vector representations.
- Use a Knowledge Graph approach with "Concept Unique Identifiers" (CUIs) from medical literature.
- Perform similarity checks to pull only relevant snippets of the EMR for the LLM to process.
- **Decision:** The client agreed to let the student team experiment with the RAG/vectorization approach to compare its efficiency against the baseline "full context" method.

3 Model Testing & Quality Assurance

- **Current Testing:** The team is currently using Gemini 1.5 Flash and Qwen (0.5B parameters) to test the framework.
- **Performance Issues:** The student team noted that while the framework works, the quality of responses from these smaller models isn't sufficient for healthcare standards.
- **Access Request:** The students requested access to the UMLS database in order to start working on Knowledge graph creation.
- **Database Access:** The client confirmed they have the database license but suggested the team apply for their own license.

4 Action Items

Action Item	Owner	Description	Deadline
RAG Comparison Report	Student Team	Comparison of token usage and latency	Next Meeting
Framework Testing	Student Team	Test RAG using vector embeddings	Ongoing
Database Access	Client Team	Provide access to UMLS dataset	Immediate
Model Benchmarking	Student Team	Compare performance across robust models	TBD

5 Next Meeting Goals

The student team will present a comparison report on the RAG implementation, specifically focusing on inference time, complexity, and token savings.